Machine Learning Approach to Fraud Detection

Mario Rodriguez, Said Ajo-Montano, Emma Jackson, Victoria Holguin
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Dr. Nogueira.

1 Introduction & Motivation

Fraud is a growing concern in banking, e-commerce, and insurance. As cybercriminals evolve, rule-based detection systems fall short. Machine learning enables real-time pattern recognition, making it a powerful tool to identify fraudulent behavior. With an estimated $48 billion lost due to digital payment fraud in 2023, organizations must implement smarter fraud detection systems. The report aims to address the challenge of fraud detection by leveraging machine learning techniques capable of adapting to evolving fraudulent behaviors. Traditional systems face limitations:

- Static rules quickly become outdated
- Manual monitoring doesn't scale
- High false positives cost billions in lost revenue

2 Exploratory Data Analysis (EDA)

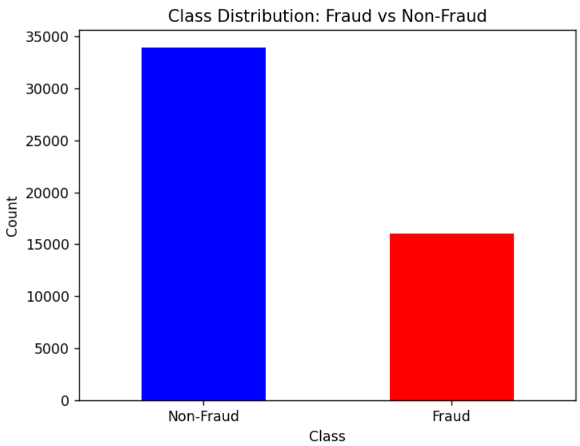
We aimed to develop a classifier $f : X \to \{0, 1\}$ that flags fraudulent transactions based on transactional and user behavioral data. We implemented two datasets, the Fraud Detection Dataset and the Credit Card Fraud Dataset. Before model development, an exploratory data analysis (EDA) was performed to better understand the structure, distribution, and characteristics of both datasets. This process included examining class distribution, temporal behavior, feature correlations, and class-separated feature distributions. Overall, EDA confirmed that the dataset required specialized handling due to class imbalance and highlighted which features provided the most discriminatory power. The findings from EDA directly influenced our pre-processing, feature selection, and model evaluation strategy.

2.1 Fraud Detection Dataset [5]

2.1.1 Class Distribution

The synthetic dataset contains approximately 6.5% fraudulent transactions, providing a more balanced target distribution than the credit card dataset (shown in Figure 1). This level of imbalance is still challenging but better supports classifier training without extensive rebalancing techniques.

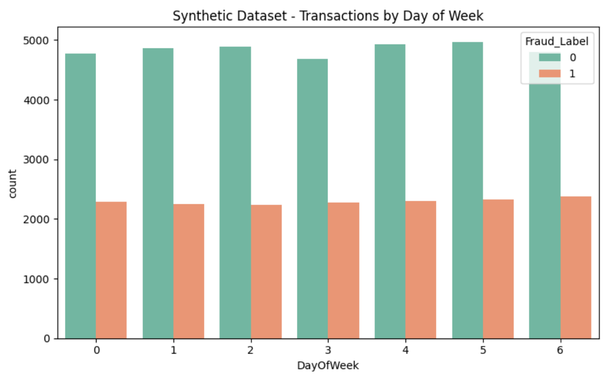
Figure 1. Moderate imbalance ink Fraud Detection Dataset



2.1.2 Daily Transaction Patterns

From the Timestamp field, we derived the day of the week. Figure 2 visualizes the distribution of transactions by day of the week, separated by fraud (label 1) and non-fraud(label 0) classes. Days are encoded from 0 (Monday) to 6 (Sunday). Fraud occurs consistently across the week, with only slight variations in volume. Non-fraudulent transactions show a relatively uniform distribution, which suggests typical user behavior is stable across all days.

Figure 2. Fraud Detection Dataset Transactions by Day of Week



2.1.3 Top Feature Histograms

We visualized the top 3 most informative numeric features by variance - Account Balance, Transaction Distance, and Average Transaction Amount - with class-separated histograms. All three plots show distinct distribution shifts between fraud and non-fraud cases: Figure 3. Transaction Distance Histogram by Class

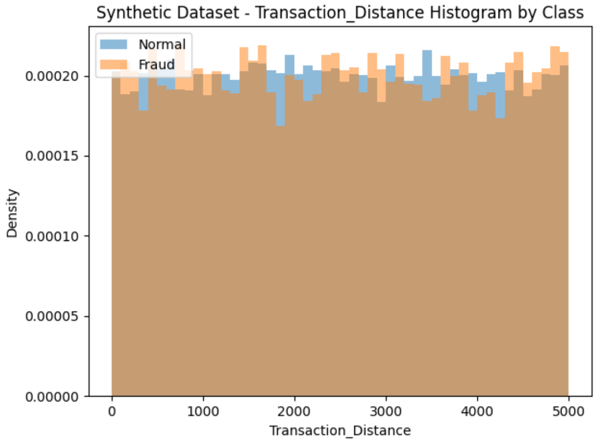


Figure 3 compares the Transaction Distance feature between normal and fraudulent transactions. While both distributions appear relatively uniform, there is a slightly higher density of fraud at the extremes, especially at longer distances. This suggests that fraud may involve transactions from atypical or distant locations, possibly indicating account compromise.

Figure 4. Account Balance Histogram by Class

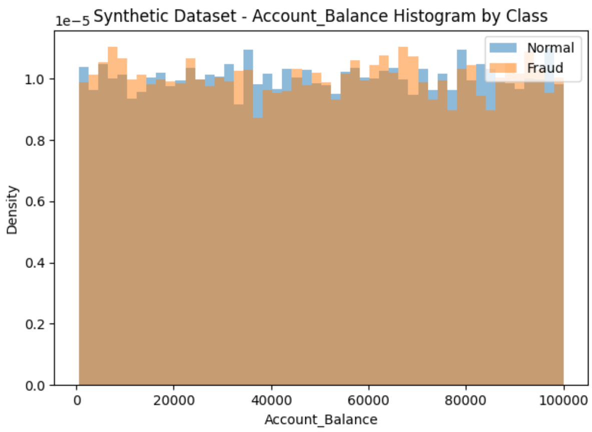


Figure 4 shows that while the distributions of normal and fraudulent transactions largely overlap, fraudulent cases skew slightly toward lower balances. This could imply that fraudsters target accounts with less security or attempt lower-value scams to avoid detection. The overall uniformity suggests that balance alone isn't decisive but may contribute as part of a feature ensemble. On its own, account balance has limited predictive power but when combined with temporal or behavioral features, it may strengthen a fraud signal.

Figure 5. Average Transaction Amount Histogram by Class

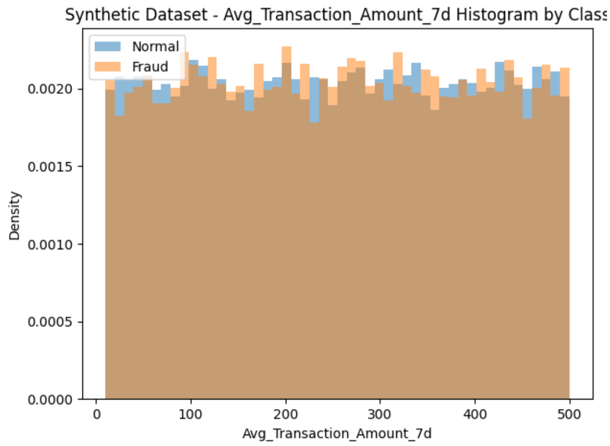


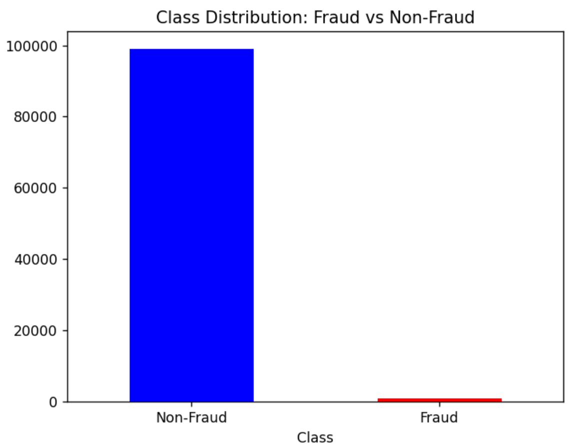
Figure 5 reveals a near-uniform distribution for both classes. However, there is a minor increase in fraud density in the middle range, possibly reflecting sudden changes in spending behavior. Since this feature captures average spending over time, it's useful for identifying deviations from a user's historical norm, which is common in fraudulent behavior.

2.2 Credit Card Fraud Dataset [1]

2.2.1 Class Distribution

The dataset is highly imbalanced, with fraudulent transactions making up only about 0.17% of the total data. This imbalance is clearly shown in the class distribution plot (Figure 6), where the majority class (non-fraud) dominates. Such skewness has serious implications for model performance, especially when using strategies or evaluation metrics like F1-score, MCC, and AUC.

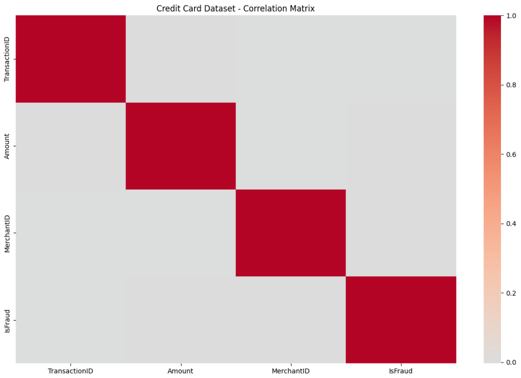
Figure 6. Extreme imbalance in the Credit Card Fraud Dataset



2.2.2 Correlation Matrix

The correlation heatmap (Figure 7) of the numerical features reveals a generally low correlation between features, suggesting minimal multicollinearity. This supports the effectiveness of ensemble methods and justifies the use of all features for modeling without immediate dimensionality reduction. However, this also implies that fraud is not linearly separable based on any single feature pair.

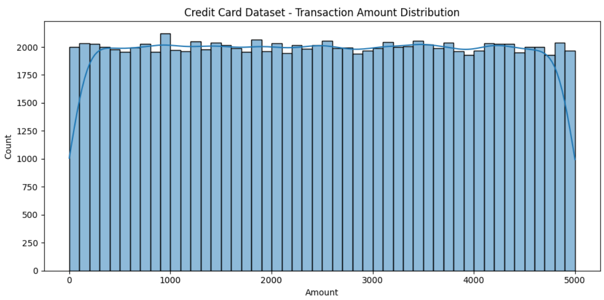
Figure 7. Credit Card Dataset - Correlation Matrix



2.2.3 Transaction Amount Distribution

Figure 8 is a Histogram of transaction amounts that shows fraudulent transactions are more common at lower numbers, while non-fraudulent transactions span a wider and higher range, This insight highlights the potential importance of transaction amount as a predictive feature and may also explain why models often achieve high recall for smaller value frauds.

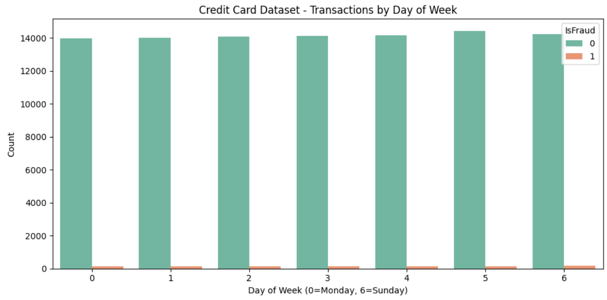
Figure 8. Transaction Amount Distribution - Credit Card Dataset



2.2.4 Day of Week Analysis

Figure count plot shows how transaction volume varies by day of the week. Non-fraudulent transactions (green bars) are evenly distributed across all days, indicating consistent usage behavior. Fraudulent transactions (orange bars) are extremely rare due to the severe class imbalance but appear evenly spread across weekdays and weekends. Temporal behavior may not drastically differ between fraud and normal activity in this dataset. However, day-of-week can still serve as a useful feature in ensemble models when paired with others, such as transaction time and amount.

Figure 9. Credit Card Transactions by Day of Week



3 Analysis Tasks

3.1 Task 1: Class Imbalance Handling

Class imbalance is a common issue in fraud detection datasets, where the number of fraudulent transactions is significantly lower than non-fraudulent ones. Here are some strategies we will be utilizing to handle class imbalance:

- **SMOTE**: Generates Synthetic samples to balance fraud and non-fraud cases. This helps balance the dataset without using simple duplication of samples.

- **Tomek Links**: Removes ambiguous samples near decision boundaries. This approach helps to better define the decision boundary between the two classes, ensuring fraud patterns aren't lost.

- **Cost-Sensitive Learning**: Adjusts model to penalize misclassifying fraud more heavily.

- **Anomaly Detection**: Isolation Forest, helps isolate anomalies by recursively partitioning the data space. One-Class SVM's supplement classification, as it learns a decision boundary that encompasses normal data points, treating outliers as fraud.

3.2 Feature Engineering & Selection

We engineer time-based, behavioral, and interaction features (e.g., transaction frequency, merchant patterns). To optimize performance:

- Use PCA, RFE, or SHAP for feature selection

- Address multicollinearity through correlation analysis or Lasso regularization

By incorporating these considerations, we can ensure that our feature engineering process enhances model accuracy, generalizability, and efficiency.

4 Model Development & Evaluation

Our solution involves preprocessing data and feature engineering, handling class imbalance, training multiple classifiers, and evaluating model performance. We test various models: including Logistic Regression, Decision Trees, Random Forest, and Gradient Boosting Machines (GBM). These models were chosen for their proven effectiveness in classification tasks and their ability to handle imbalanced datasets, which is common in fraud detection scenarios.

However, we recognize the importance of considering deep learning approaches such as Long Short-Term Memory (LSTM) networks, Convolutional Neural Networks (CNNs), and Autoencoders. These models are particularly suited for processing sequential data, which is relevant for fraud detection as it often involves temporal transaction data. LSTMs and CNNs can capture complex patterns and dependencies over time, potentially improving the accuracy of fraud detection.

Additionally, ensemble methods like bagging and boosting have been shown to outperform single models by up to 12% (as stated in our related works section). Therefore, we will also explore advanced ensemble techniques such as XGBoost and stacking models. These methods combine the strengths of multiple models, leading to better performance and robustness in detecting fraudulent transactions. By incorporating both traditional machine learning models and deep learning approaches, along with powerful ensemble methods, our team aims to develop a comprehensive and effective fraud detection system.

4.1 Hyperparameter Tuning

We use GridSearchCV offered by Scikit-learn to find optimal hyperparameters for the models. It trains each model using a variety of parameters to determine which ones give the best results.

4.2 Model Evaluation

We evaluate our models using standard metrics such as precision, recall, and F1-score provide foundational insight into the model's ability to correctly identify fraudulent activity without generating excessive false positives. However, because fraud datasets are typically skewed, we also rely on the Area Under the Receiver Operating Characteristic Curve (AUC-ROC) to assess the model's ability to distinguish between the classes across various thresholds.

Beyond these conventional metrics, we incorporate the Matthews Correlation Coefficient (MCC), which is particularly useful for imbalanced datasets, offering a balanced evaluation that takes into account true and false positives and negatives. To further evaluate business utility, we apply lift analysis, which quantifies how much better the model performs compared to random selection, especially in the context of resource-constrained investigations.

Finally, we employ threshold tuning to strike a practical balance between precision and recall, ensuring that our model aligns with cost-sensitive environments where missing a fraud case is far more expensive than a false alert.

4.3 Efficiency

We look at the overall runtime of our code as well as the time taken by each model to determine how efficient our solution is. The total runtime for our solution was approximately 8 seconds (8.0207), with the Support Vector Machine being the slowest at 3.722 seconds. Logistic regression performed the fastest, with a time of 0.0088 seconds. The time for other models ranges from 0.0789 seconds to 0.1863 seconds. This tells us that the other models are all relatively close together in terms of efficiency. The running time of the models adds up to about 4 seconds, which means roughly half of the time is dedicated to training the models, and the remaining time is spent in preprocessing and data visualization.

5 Adversarial Robustness

Modern fraud detection systems must be resilient against deliberate manipulation by adversaries who seek to bypass detection. These systems are vulnerable to a variety of attacks, including data poisoning, where the training data is subtly corrupted to mislead the model, model extraction, where attackers probe the system to replicate its behavior, and evasion attacks, where inputs are crafted to avoid triggering fraud flags.

To counter these threats, several defense strategies can be implemented. Static model deployment, where model updates are less frequent and subject to rigorous validation, reduces the exposure to ongoing manipulation. Output sanitization, such as withholding confidence scores, limits the information available to adversaries attempting model reverse-engineering. Query filtering mechanisms can detect and block suspicious input patterns that resemble probing behavior.

In addition, adversarial training—which involves incorporating maliciously crafted examples into the training process—can improve the model's resilience to subtle manipulations. By anticipating and preparing for adversarial tactics, we aimed to develop a more secure and trustworthy fraud detection system capable of adapting to evolving threats.

6 Relation Work & Insights

Several existing works have shaped our understanding of effective strategies in fraud detection. The **Fraud Dataset Benchmark (FDB)** [4] highlights the clear advantage of ensemble models, reporting 12% improvement in AUC over single-model approaches. This informed us that there should be some focus on ensemble techniques for our fraud detection experiments.

From an operational standpoint, **DataDome** demonstrates the power of high-dimensional feature engineering and real-time inference by leveraging over 150 features to process more than 5 million requests per second with 99.998% precision. It offers a real-world example of scalable, production-grade fraud detection. This informs our design choices around model complexity and performance evaluation.

To emulate realistic fraud behavior, we also looked into synthetic datasets like **PaySim** [3] **and Sparkov** [2]. These generators model temporal and contextual patterns observed in the real world, such as 73% of fraud occurring at night and targeting high-risk merchants like electronics, which helps in validating the behavioral relevance of our chosen datasets and performance metrics.

7 Conclusion

This project presents a comprehensive approach to detecting fraudulent transactions by leveraging a wide array of machine learning models and comparing their performance across both real-world and synthetic datasets. Our motivation stemmed from the urgent need to mitigate substantial financial losses incurred due to credit card fraud and the challenges posed by imbalanced data distributions and subtle behavioral patterns that traditional rule-based systems often fail to detect.

Through extensive experimentation, we evaluated models ranging from classic algorithms such as Logistic Regression and Decision Tress to more sophisticated architectures like XGBoost, Stacked Ensembles, Convolutional Neural Networks (CNNs), and Long Short-Term Memory Networks (LSTMs). We also incorporated Autoencoders as an unsupervised method to learn underlying patterns of normal behavior.

Our results show that the ensemble models (e.g., XGBoost and stacking) and GBM consistently outperformed others in terms of F1 Score, AUC and MCC, reflecting their robustness in identifying instances of minority class. Deep learning methods, particulary CNNs and LSTMs, demonstrated reasonable predictive performance but required significantly more training time and hyperparameter tuning. Autoencoders, while less accurate, proved valuable in learning data representation and highlighting anomalies.

In addition to model performance, the use of synthetic data allowed us to validate the generalizability of our approach. Although synthetic datasets led to slightly reduced performance across most metrics, the trend of relative model effectiveness remained consistent. This confirms that the chosen features and architectures retain predictive power even under controlled data generation conditions.

Overall, this project illustrates the value of combining domain-informed feature engineering, proper handling of class imbalance, and diverse modeling techniques to build a robust fraud detection pipeline. Future work could involve deploying these models in real-time systems, incorporating temporal patterns in a multi-sequence setup, and exploring federated learning for enhanced privacy and data sharing.

References

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